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Ca^{2+} signaling–mediated low-intensity pulsed ultrasound–induced proliferation and activation of motor neuron cells

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ABSTRACT

Motor neuron diseases (MND) including amyotrophic lateral sclerosis and Parkinson disease are commonly neurodegenerative, causing a gradual loss of nerve cells and affecting the mechanisms underlying changes in calcium (Ca^{2+})-regulated dendritic growth. In this study, the NSC-34 cell line, a population of hybridomas generated using mouse spinal cord cells with neuroblastoma, was used to investigate the effect of low-intensity pulsed ultrasound (LIPUS) as part of an MND treatment model. After NSC-34 cells were seeded for 24 h, LIPUS stimulation was performed on the cells at days 1 and 3 using a non-focused transducer at 1.15 MHz for 8 min. NSC-34 cell proliferation and morphological changes were observed at various LIPUS intensities and different combinations of Ca^{2+} channel blockers. The nuclear translocation of Ca^{2+}-dependent transcription factors was also examined. We observed that the neurite outgrowth and cell number of NSC-34 significantly increased with LIPUS stimulation at days 2 and 4, which may be associated with the treatment's positive effect on the activation of Ca^{2+}-dependent transcription factors, such as nuclear factor of activated T cells and nuclear factor-kappa B. Our findings suggest that the LIPUS-induced Ca^{2+} signaling and transcription factor activation facilitate the morphological maturation and proliferation of NSC-34 cells, presenting a promising noninvasive method to improve stimulation therapy for MNDs in the future.

1. Introduction

Motor neuron diseases (MNDs) are a group of neurodegenerative diseases caused by aging, inflammation, environmental pollutants, nutritional factors, or an overproduction of free radicals that attack nerve cells [1–5] and that induce neuronal cell death due to neurological disorders and oxidative stress [6,7]. However, there is no stimulation therapy that currently exists for preventing the progression of MNDs [8], except a handful of drugs such as Edaravone [9], Ezogabine [10], and dimethyl fumarate [11] that can be used to decrease the overactivation of neurons, increase motor neuron survival, and increase the number and function of neuronal cells. In addition, electric electrodes have been used to attenuate oxidative stress in MNDs [12]. The formation of functional neurons involves the process of differentiation, in which axons and dendrites determine the appropriate synaptic partners for neurotransmitter activity and extend toward the target cells [13]. The dysregulation of neuronal growth and development is a key process in the progressive degeneration of neurons [14].

In previous studies, morphological changes in neurons mediated by different signaling pathways have depended on the precisely regulated polymerization, convergence and crosslinking of cytoskeletons. However, several transcription factors, namely *cAMP-response element binding* protein (CREB), nuclear factor of activated T cells (NFAT), nuclear factor-kappa B (NFκB), p53, c-Jun, activating transcription factor 3, signal transducer and activator of transcription 3, SOX11, SnoN, and Krüppel-like factors (KLFs), have been demonstrated to influence neural development and axon regeneration [15]. Transcription factors present in developing dendritic spines are locally translated, phosphorylated, and transported to the nucleus [16]. Calcium ion (Ca^{2+}) is a highly versatile secondary messenger that plays a key and pervasive role in intracellular signal transduction and the regulation of many cellular functions, including proliferation, development, differentiation,

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migration, transcription factor activation, and apoptosis, in neurons [17,18]. Initially, Ca^{2+} transmits electrical signals down the axons to activate the synaptic terminals and then metabolizes neurotransmitters into synapses. Therein, Ca^{2+} influx would mediate elevation of local membrane potential that contributes to generating an action potential. And action potential induced Ca^{2+} influx is essential for neurotransmitter release in the axonal terminal. Therefore, most genes transcriptions are related to Ca^{2+}-dependent transcription factors. Ca^{2+} signaling is mediated mainly by voltage-sensitive Ca^{2+} channels (VSCCs) and N-methyl-D-aspartate receptors (NMDAs) in excitatory neurons. Many Ca^{2+}-dependent transcription factors, including NFAT, NFκB, and CREB, are likely to be activated during signaling [19–21]. For example, the intra-axonal translation of CREB mRNA may affect the phosphorylation of nuclear CREB, leading to an increase in neuron survival [22]. An increase in Ca^{2+} in response to synaptic activity or depolarization in nerve cells leads to the activation of calcineurin phosphatase, which causes the dephosphorylation of NFATs for its translocation to the cell nucleus [23,24]. NFκB is a downstream modulator of neurite development for Fas apoptosis inhibitory molecule signaling [15,25].

Currently, several noninvasive methods of mechanical stimulation, such as high-intensity or low-intensity pulsed ultrasound (LIPUS,) electrical, optical, and magnetic approach, have been proposed for motor neuron stimulation. Among them, LIPUS has been reported to be able to stimulate targets at deep distances in cells, whether in animals or humans. LIPUS may have therapeutically relevant efficacy and did not cause cytotoxicity and inflammatory response on growing cells to different stimuli of in vitro cell culture [26]. In 1950, Fry et al. demonstrated the ability of the high-intensity pulsed ultrasound at 1 MHz and a spatial average temporal average intensity (I_{SATA}) of 30 mW/cm^2 to regulate neural activity in the ventral nerves of crayfish and to prevent light-evoked activation in their visual cortices [27]. LIPUS was used to treat the osteoporosis and osteoporotic bone defect in ovariectomized rats [28]. Crisci et al. applied LIPUS at 1.5 MHz with a I_{SATA} of 16 mW/cm^2 to accelerate peripheral neural tissue regeneration in rats [29]. In addition, Schwann cells stimulated with an ultrasound I_{SATA} of 27–50 mW/cm^2 at 1 MHz exhibited an increased proliferation and induction of neurite outgrowth [30,31]. LIPUS with a peak negative pressure of 0.42 MPa [32] and a spatial peak temporal average intensity (I_{SPTA}) of 360 mW/cm^2 [33] can excite postexposure neurite outgrowth and significantly increase the density of dendritic spines in rats. The changes and homeostasis of intracellular ion concentration play a very important role in the normal physiological function and pathological progress of cells. Several studies have been reported that ultrasound can activate several mechanically sensitive ion channels by acoustic radiation [34–36]. Induction of the touch receptor neurons through the MEC-4 ion channel was observed in *Caenorhabditis elegans* by an ultrasonic transducer at 300 kHz [34]. It was also demonstrated to activate neurons via the mechanical-induced ASIC1a channel in mouse brain at 10 mW/cm^2 (I_{SATA}) ultrasound stimulation [35]. In addition, a tone burst ultrasound wave was generated by a focused transducer (10 MHz, 0.3–4.9 W/cm^2 of I_{SPTA}) that can modulate the currents flowing through the ion channels [36]. Furthermore, LIPUS can be used to stimulate electrical activity in neurons, trigger TTX-sensitive neuronal activity, and induce brain activity; studies have reported that LIPUS about 0.5 MHz can significantly change the activity of approximately 40% of neurons on hippocampal slice cultures of mouse, monkey, and human brains [37–39]. Moreover, LIPUS may modulate many cell activities via regulated Ca^{2+} signaling [40–44], such as inducing intracellular Ca^{2+} dynamics in HIT-T15 pancreatic β-cells [40]; activating mechanosensitive, and voltage-gated Ca^{2+} channel TRPC1 in kidney and skeletal muscle [41] and promoting angiogenesis via the AKT signaling pathway and DNA methylation in endothelial cells [42]. For ultrasonic neuromodulation, focused ultrasound (15 W/cm^2) also causes the opening of specific Ca^{2+}-selective mechanosensitive ion channels in primary cortical neurons [43]. Transducer emitting ultrasonic waves (0.44 MPa of peak pressure) transiently increases intracellular Ca^{2+} in individual neurons in awake mouse brains [44]. Especially, LIPUS activates TrkB-Akt and Ca^{2+}-CaMK signaling pathway at 1 MHz with an I_{SPTA} of 110 mW/cm^2 [21] in astrocyte and induces Ca2 + influx at 0.5 MHz with a peak pressure of 0.3 MPa via the mechanosensitive ion channels MscL-G22S in different neurons of mouse brain [45].

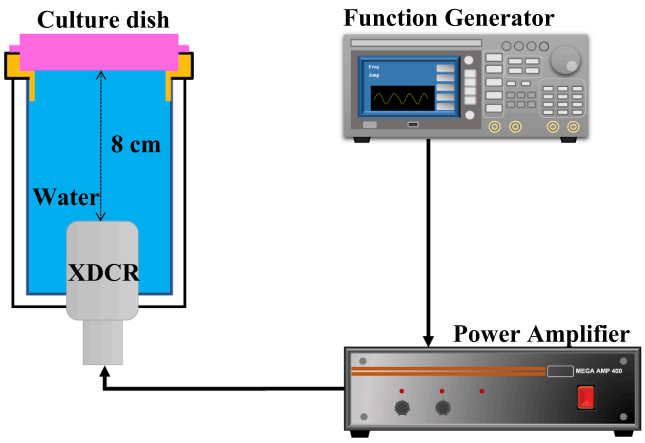


Fig. 1. Schematic of ultrasound stimulation system. The LIPUS signal was created using a function generator and an RF amplifier and stimulated each sample in a culture dish using a non-focused transducer at 1.15 MHz through an acrylic tube containing 8 cm high of water. The LIPUS was administered in two types of pulse duration (200, and 500 cycles, respectively) and at different peak-to-peak voltages (1, 3, and 5 V_{pp}) for 8 min at a center frequency (f_c) of 1.15 MHz and a pulse repetition frequency (PRF) of 1 kHz.

The aforementioned results suggest that certain Ca^{2+}-dependent transcription factors can enhance the function and viability of motor neurons as an outstanding approach to the treatment of MNDs [15–25] and that LIPUS with different ultrasound parameters can be used to influence Ca^{2+} signaling and the morphology and. proliferation of neurons [21,30–37]. However, little has been reported about the effect of LIPUS stimulation on motor neuron cell lines and the effects of LIPUS on the activation of downstream Ca^{2+} signaling proteins, such as the Ca^{2+}-dependent transcription factors NFAT and NF κ B. Therefore, the development of a LIPUS therapy that can treat or support the treatment of MNDs is essential. In this study, we investigated the effects of LIPUS at various intensities to determine whether LIPUS could activate the growth and differentiation of NSC-34 motor neurons through the induction of Ca^{2+} signaling.

2. Materials and methods

2.1. Cell culture

NSC-34, a mouse motor-neuron-like hybrid cell line, was originally produced using cells from a mouse spinal cord with neuroblastoma [46,47]. NSC-34 cells were maintained in high glucose Dulbecco's modified Eagle's medium (DMEM, #DMP08, Caisson) supplemented with 10% fetal bovine serum (FBS; #2156016, Gibco) and 1% penicillin-streptomycin and incubated at 37 °C with 5% CO_2.

2.2. LIPUS stimulation

As illustrated in Fig. 1, a non-focused transducer (A314S, Olympus, Waltham, MA, USA) with center frequency of 1.15 MHz and aperture size of 0.75 in. was driven by an electrical driving signal from a function generator (AFG 3022B, Tektronix, Irving, TX, USA) and a radio frequency (RF) amplifier (25A250A, Amplifier Research, Souderton, PA, USA) and then the generated ultrasound waves were used to stimulate the NSC-34 cells through an acrylic tube containing water. The NSC-34 cells (5×10^4) were seeded in 3 cm plastic dishes. The cells were divided into two groups: one did not undergo LIPUS stimulation (control group)

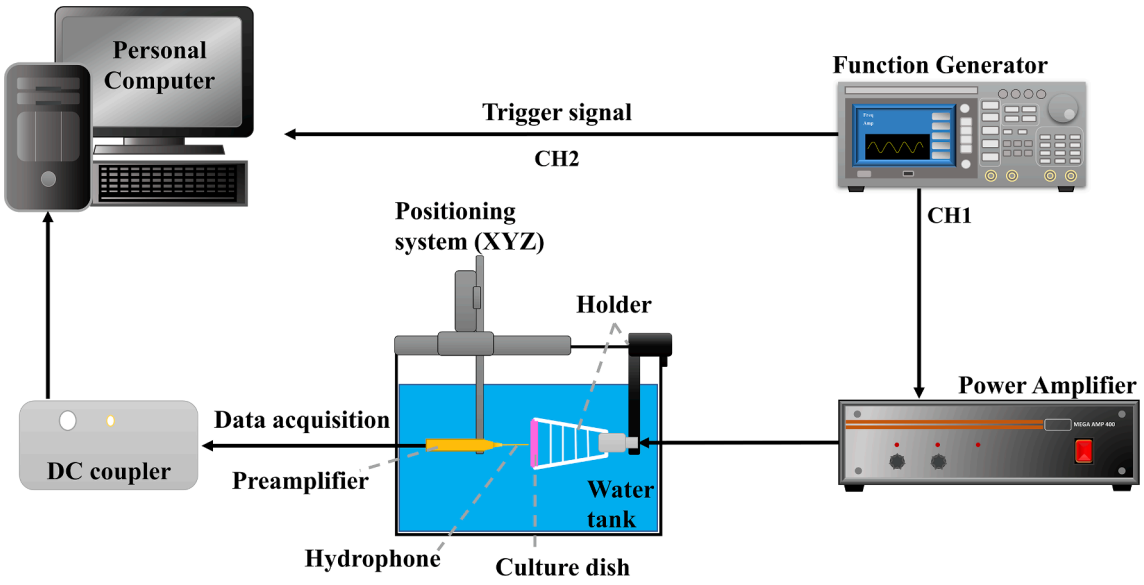


Fig. 2. Schematic of a hydrophone system for measuring the acoustic outputs of transducer.

and the other one underwent LIPUS stimulation (LIPUS group). The LIPUS with the spatial peak average temporal average intensity (I_{SPTA}) that ranged from approximately 141 to 1267 mW/cm^2 was applied to stimulate the LIPUS group cells.

Acoustic outputs were measured by a hydrophone system that contains a hydrophone needle (SN 2392, Precision Acoustics, Dorchester, Dorset, UK) with a diameter of 0.2 mm, a preamplifier (PA 16039, Precision Acoustics, Dorchester, Dorset, UK) and a DC Coupler (DCP S556, Precision Acoustics, Dorchester, Dorset, UK), as shown in Fig. 2. The hydrophone needle was located at the midpoint in front of the surface of the culture plate and slowly scanned to take acoustic pressure measurements across the transverse and longitudinal directions using an XYZ motor stage. The measured voltage values were converted to pressure values via the corresponding calibration table in the hydrophone calibration panel according to formula (1):

$$I = \frac{P^2}{\rho c} \tag{1}$$

where the P is pressure, I is intensity, and ρ is the density of water (1000 kg/m^3), and c is the sound velocity in water (1500 m/s) [48]. Finally, the I_{SPTA} was calculated, which can be obtained by averaging I (t) over N cycles from equation (2):

$$I_{SPTA} = \frac{P_{max}^2}{2\rho c} \times \frac{N \times PRF}{f_c} \tag{2}$$

where, f_c is the center frequency and PRF represents the pulse repetition frequency [32].

2.3. Morphological observation

The LIPUS stimulation was performed after cells had been seeded for 24 h. Subsequently, the medium in each dish was refreshed (DMEM, 5% FBS, and 1% P/S). This procedure was repeated on day 3, and the morphological changes were observed under an inverted research microscope (IX-71, Olympus Corporation, Shinjuku, Tokyo, Japan) on days 2 and 4. The NSC-34 cells were treated using a Ca^{2+}-free medium (#1684796, Gibco), and two Ca^{2+} channel blockers: 50 μM of gadolinium (Gd^{3+}; #439770, Sigma-Aldrich) and 200 μM of lanthanum (La^{3+}; #262072, Sigma-Aldrich) for 30 min before LIPUS stimulation to verify that the Ca^{2+} signaling was induced by the LIPUS.

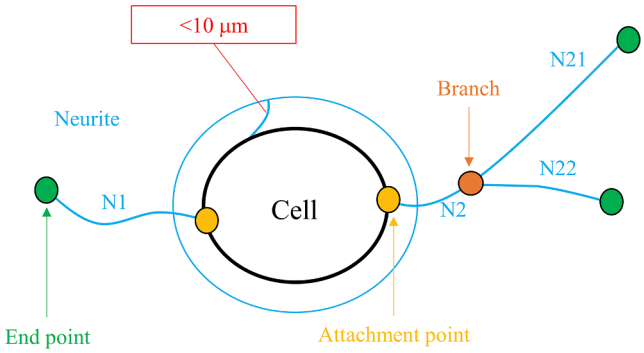


Fig. 3. Overview of quantification of neurite outgrowth. Neurite (blue color) outgrowth is a neural morphological development process. Neurites shorter than 10 μm were not analyzed in this study. Attachment points (yellow) were determined by their positional adhesives to the cell body (black). The end points (green) are the end positions of each neurite. Branches (orange) are defined by the division of the primary neurite (N2) into two secondary neurites (N21 and N22).

2.4. Quantification of neurite outgrowth

The neurite outgrowth of the NSC-34 cells was determined by measuring the number and average length of the neurites and the mean number of attachment and end points per cell (Fig. 3). All analyses were conducted by Fiji/Image J 2.0 software (National Institutes of Health, Bethesda, MD, USA).

2.5. Viability assay

The NSC-34 cells were treated with LIPUS on days 1 and 3. The cytotoxic effect of the LIPUS was measured by labeling dead cells with Ethidium (#E1169, Invitrogen) and all nuclei with Hoechst 33,342 (#HO 33342, Gibco) on day 4. The NSC-34 cells were incubated with the Ethidium and Hoechst 33,342 at 37 °C for 30 min. Thereafter, fluorescence images were taken under an inverted research microscope (IX-71, Olympus Corporation, Shinjuku, Tokyo, Japan).

2.6. Cell proliferation assay

The NSC-34 cells were stimulated using LIPUS every 2 days. At day 5,

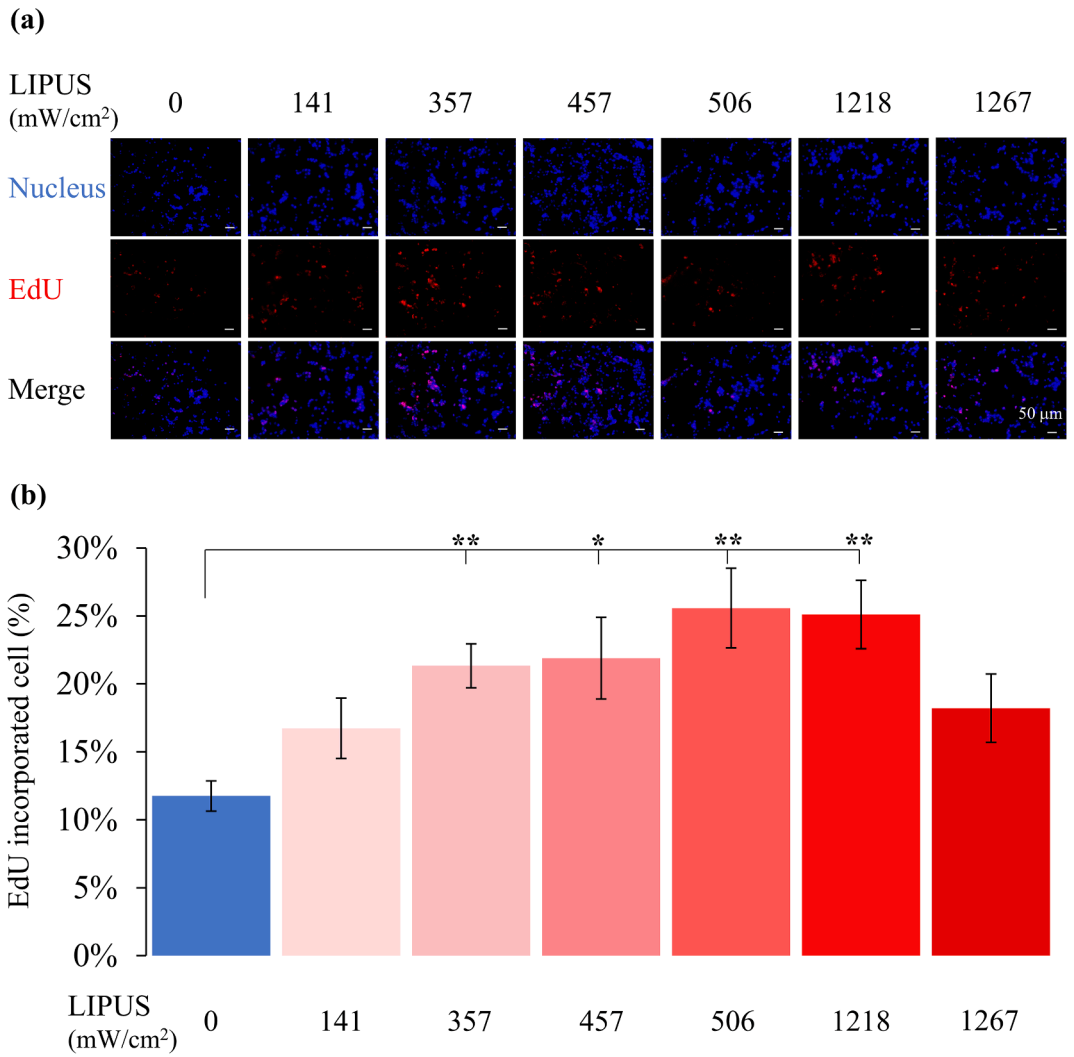


Fig. 4. Effects of LIPUS on the proliferation of NSC-34 cells at various intensities levels. A total of 5×10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with LIPUS at various intensities (control group: 0 mW/cm^2; LIPUS groups: 141 mW/cm^2, 357 mW/cm^2, 457 mW/cm^2, 506 mW/cm^2, 1218 mW/cm^2, and 1267 mW/cm^2) on days 1 and 3; 24 h after the end of LIPUS stimulation, the cells were co-incubated with 1μ g/mL Hoechst 33,342 and 10 μM EdU for 4 h. (a) The fluorescence images of nuclei (blue) and EdU (red) in NSC-34 cells. Each image was taken using an inverted microscope. Scale bar: 50 μm. (b) Cell proliferation is presented as the percentage of EdU-incorporated NSC-34 cells. Each bar represents the mean ± SEM from three independent experiments (10 pictures were taken for each condition in every experiment). *$p < 0.05$, **$p < 0.01$.

the cells were incubated with EdU (5-ethynyl-2′-deoxyuridine, 10 μM), a thymidine nucleoside analog, at 37 °C for 4 h to track the DNA during DNA synthesis. Subsequently, cells were fixed in 4% paraformaldehyde (PFA; #30525–89-4, Sigma-Aldrich) for 15 min, blocked with bovine serum albumin (BSA; #0948–46-8, UniRegion Bio-Tech) for 30 min, and permeabilized in 1 mL of 0.5% Triton X-100 (#A9778, PanReac AppliChem) for 20 min. Thereafter, cells were blocked with BSA again, and 0.5 mL of Click-iT reaction cocktail was added for EdU detection before DNA labeling was performed for 30 min. Subsequently, the cells were rinsed with 1 mL of BSA and phosphate buffered saline (PBS). Finally, the cells were stained with the nuclear dye Hoechst 33342 (1 μg/mL) and the Alexa Fluor 594 azide in a Click-iT EdU imaging kit (#C10339, Invitrogen) for 30 min. EdU-positive images were taken under an inverted research microscope (IX-71, Olympus Corporation, Shinjuku, Tokyo, Japan).

2.7. Immunofluorescence staining

A Ca^{2+}-free medium and two Ca^{2+} channel blockers, Gd^{3+} (50 μM) and La^{3+} (200 μM), were applied for 30 min before LIPUS stimulation to examine the activation of Ca^{2+}-dependent transcription factors by LIPUS. After LIPUS stimulation, NSC-34 cells were immediately fixed in 4% PFA for 15 min before being incubated in the permeabilizing buffer (1X PBS, 5% fetal bovine serum, and 0.3% Triton X-100) for 10 min. The cells were blocked with an ultra-pure commercial BSA for 1 h; the primary antibody was then added to the cells, and each dish was covered with a parafilm membrane at 4 °C overnight. After the unbound primary antibody was removed using PBS, a fluorophore-conjugated secondary antibody was added to the cells for 1 h at room temperature, and the mixture was placed in a box to shield it from light. Again, the unbound secondary antibody was removed by rinsing with PBS for 50 min. Finally, confocal images were taken under a multiphoton laser scanning microscope (MPE, Olympus Corporation, Shinjuku, Tokyo, Japan). The primary and secondary antibodies used in this study were anti-NFκB p65 (#C22B4, Cell Signaling), anti-NFAT1 (#bs-4246R, Bioss), anti-phospho-CREB (#9198S, Cell Signaling), and Alexa 488-conjugated goat antirabbit antibody (#A11070, Invitrogen), 2 μM of Ionomycin (#I0634, Sigma-Aldrich) and 1 μM of phorbol 12-myristate 13-acetate (PMA; #P8139, Sigma-Aldrich) were used as positive controls for NFAT and NFκB, respectively.

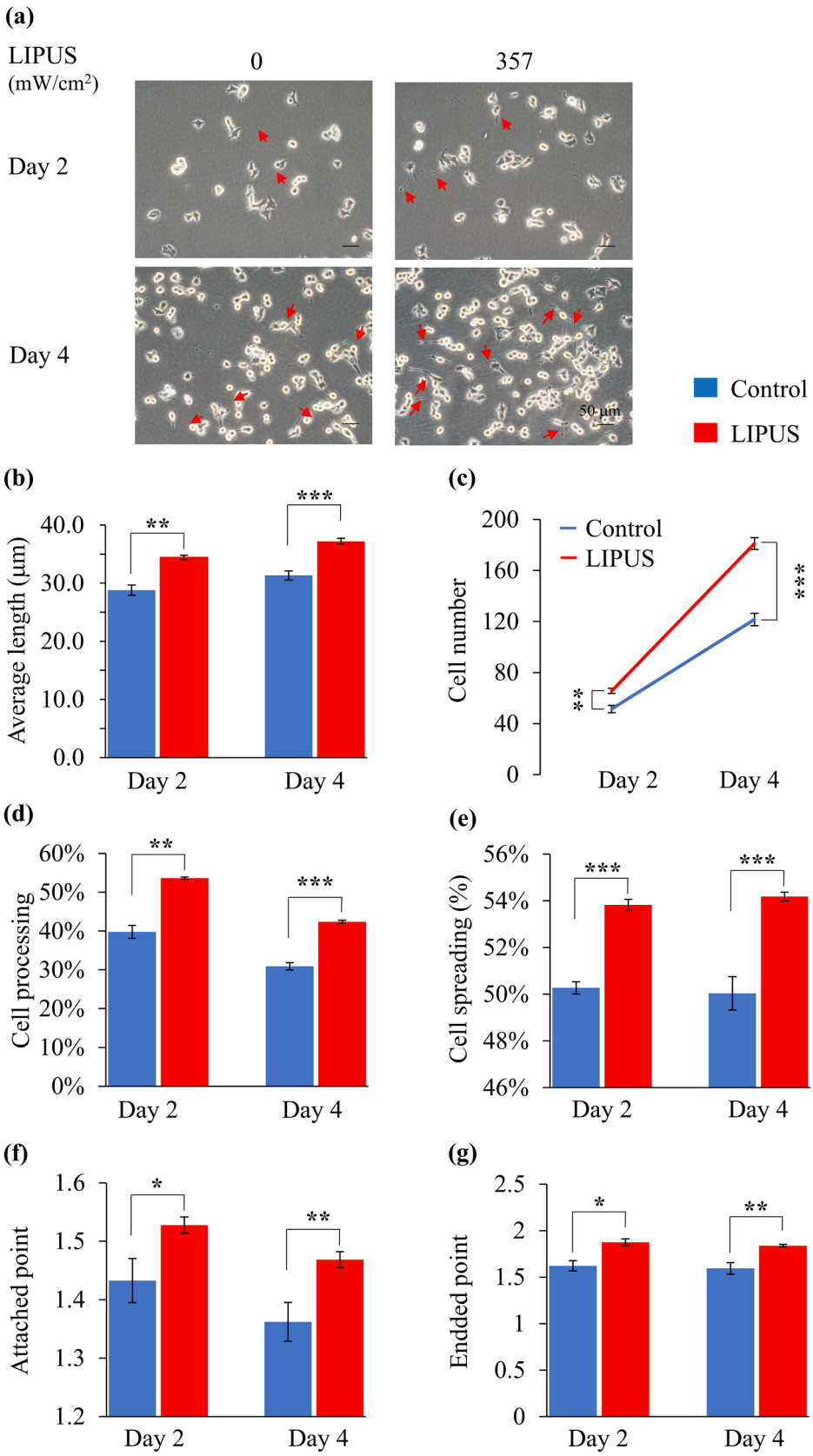


Fig. 5. Morphological changes of NSC-34 cells induced by LIPUS stimulation. A total of 5×10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. The morphological changes of NSC-34 cells were observed under an inverted phase-contrast microscope on days 2 and 4. (a) Neurite outgrowth was observed under an inverted phase-contrast microscope and is indicated by the red arrows in the pictures. Scale bar: 50 μm. Quantitative results for (b) the average length of cell processes, (c) the number of NSC-34 cells, (d) the percentage of NSC-34 cells undergoing processes, (e) the percentage of cell spreading, (f) the attachment points, and (g) the end points in the LIPUS group were significantly higher than those of the control group. Each bar represents the mean ± SEM from three independent experiments (10 pictures were taken for each condition in every experiment). $^{**}p < 0.01$; $^{***}p < 0.001$.

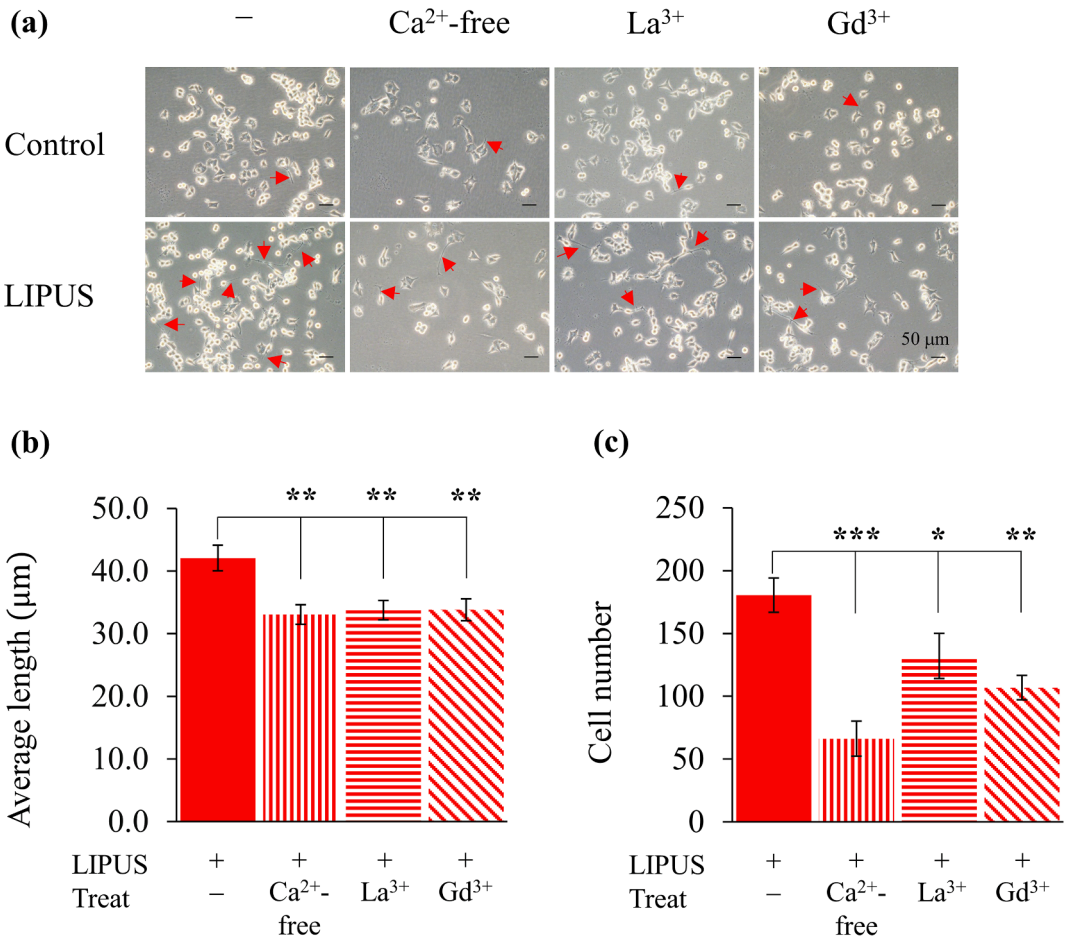


Fig. 6. Ca^{2+} signaling is involved in the LIPUS-induced morphological changes and proliferation of NSC-34 cells. A total of 5×10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. Ca^{2+} channel blockers (200 μM La^{3+} and 50 μM Gd^{3+}) and a Ca^{2+}-free medium were administered for 30 min before LIPUS stimulation. The morphological changes of the NSC-34 cells were observed under an inverted phase-contrast microscope on day 4. (a) The neurite outgrowth was observed under an inverted phase-contrast microscope and is indicated by the red arrows in the pictures. Scale bar: 50 μm. Quantitative analysis of the (b) average length of neurite per cell, and (c) number of cells. Each bar represents the mean ± SEM from three independent experiments (5 pictures were taken for each condition in every experiment). *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

2.8. Statistical analysis

All data are expressed in terms of the mean ± standard error of the mean (SEM) of three independent measurements. A Student's *t* test was used to compare the control and experimental groups. A p value < 0.05 was considered statistically significant.

3. Results

3.1. Effect of LIPUS on cell proliferation

Many studies have reported the cellular effects of LIPUS at a variety of settings [21,30–36,49]. In the present study, different LIPUS intensities (I_{SPTA}) of 141 mW/cm^2, 357 mW/cm^2, 457 mW/cm^2, 506 mW/cm^2, 1218 mW/cm^2, and 1267 mW/cm^2 were applied for 8 min on days 1 and 3. The EdU staining kit was used to quantify the effect of the LIPUS parameters on NSC-34 cell proliferation. NSC-34 nuclei stained with DNA dye Hoechst 33,342 exhibited blue fluorescence, whereas the proliferating nuclei incorporated with EdU was indicated by their red fluorescence (Fig. 4a). The percentages of EdU-incorporated cells at each of the mentioned LIPUS intensities were 16.7% ± 2.2%, 21.3% ± 1.6%, 21.9% ± 3.0%, 25.6% ± 2.9%, 25.1% ± 2.5% and 18.2% ± 2.5% respectively, compared with 11.8% ± 1.1% in the control group (Fig. 4b). These results indicate that LIPUS stimulation promoted cell proliferation, especially at 357 mW/cm^2. The low intensity of LIPUS can not only minimize damage to cells but also significantly stimulate NSC-34 cell proliferation. Therefore, the LIPUS intensity at 357 mW/cm^2 was selected for use in the following procedures.

3.2. LIPUS on differentiation of NSC-34 cells

The optimal LIPUS parameters, as indicated in Fig. 4, were applied to explore the effect of LIPUS on the morphological and developmental changes of NSC-34 cells after 24 h LIPUS treatment at days 2 and 4. As indicated by the red arrows in Fig. 5a, the neurites were particularly elongated under LIPUS stimulation. The mean length of the LIPUS group (day 2: 34.42 ± 0.4 μm; day 4: 37.21 ± 0.5 μm) increased significantly more than that of the control group (day 2: 28.79 ± 0.9 μm; day 4: 31.32 ± 0.8 μm; Fig. 5b). The number of NSC-34 cells in the LIPIUS group (day 2: 65.67 ± 2.0 cells; day 4: 181.10 ± 4.6 cells) also increased significantly more than that of the control group (day 2: 51.40 ± 2.8 cells; day 4: 121.60 ± 4.9 cells; Fig. 5c). Furthermore, the levels of other morphological characteristics, such as those pertaining to attachment points, end points, cell with outgrowth processes, and cell spreading, also increased significantly more in the LIPUS group than in the control group. The percentage of NSC-34 cells with cell processes was significantly higher in the LIPUS group (day 2: 53.6% ± 3.0%; day 4: 42.40% ± 0.4%) than in the control group (day 2: 39.8% ± 1.7%; day 4: 30.90% ± 0.9%; Fig. 5d). The percentage of NSC-34 cells spreading was higher in the LIPUS group (day 2: 53.8% ± 0.2%; day 4: 54.20% ± 0.2%) than in

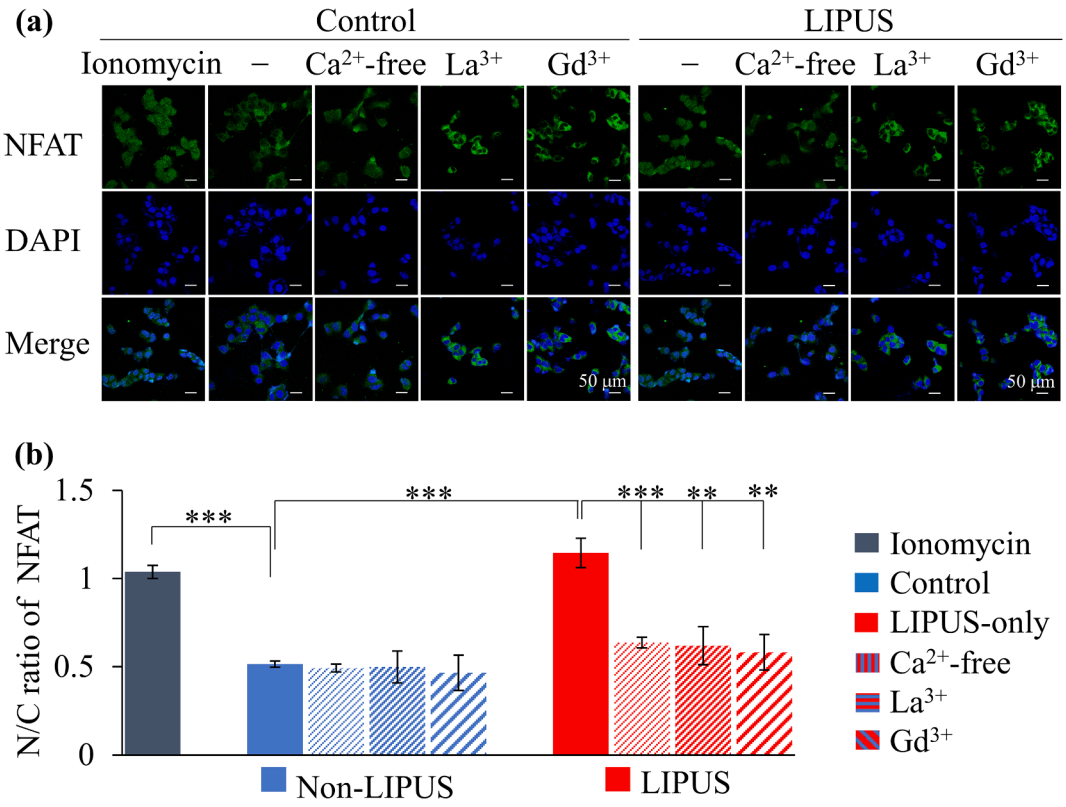


Fig. 7. LIPUS stimulation induces activation of transcription factor NFAT in NSC-34 cells. A total of 5×10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. Ca^{2+} channel blockers (200 μM La^{3+} and 50 μM Gd^{3+}) and a Ca^{2+}-free medium were administered for 30 min before LIPUS stimulation. Cells in the positive control groups were treated with 2 μM ionomycin for 30 min. (a) Immunofluorescence staining of NFAT (green) and nuclei (blue); images were taken using confocal microscopy. Scale bar: 50 μm. (b) Quantitative analysis of the nucleus-to-cytosol ratio for NFAT fluorescence intensity. Each bar represents the mean ± SEM from three independent experiments (5 pictures were taken for each condition in every experiment). $^{**}p < 0.01$; $^{***}p < 0.001$.

the control group (day 2: 50.3% ± 0.3%; day 4: 50.00% ± 0.7%; Fig. 5e). The mean numbers of attachment points (Fig. 5f) and end points (Fig. 5g) of the LIPUS group (day 2: 1.53 ± 0.02 attachment points and 1.88 ± 0.04 end points; day 4: 1.47 ± 0.01 attachment points and 1.84 ± 0.01 end points) increased more than those of the control group (day 2: 1.43 ± 0.04 attachment points and 1.62 ± 0.06 end points; day 4: 1.36 ± 0.03 attachment points and 1.59 ± 0.06 end points).

3.3. Ca^{2+} signaling in LIPUS-induced morphological changes and proliferation of NSC-34 cells

The morphological changes and proliferation of the NSC-34 cells were observed under LIPUS stimulation. We proposed that LIPUS-induced Ca^{2+} signaling was involved in these changes. Thus, a Ca^{2+}-free medium and two Ca^{2+} channel blockers, La^{3+} and Gd^{3+}, were used to demonstrate the effect of Ca^{2+} signaling on the induction of neurite outgrowth and cell proliferation. As illustrated in Fig. 6a, neurite outgrowth was observed among fewer of the NSC-34 cells in the Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups than of those in the LIPUS-only group. Quantitative analysis indicated that the mean neurite length of the control LIPUS group (42.08 ± 2.03 μm) was significantly longer than those of the groups with inhibited Ca^{2+} signaling (Ca^{2+}-free: 33.06 ± 1.58 μm, La^{3+}: 33.75 ± 1.54 μm, Gd^{3+}: 33.83 ± 1.75 μm; Fig. 6b). The number of cells was also slightly lower in La^{3+} and Gd^{3+} stimulation groups (132.12 ± 17.90 cells and 106.92 ± 9.82 cells, respectively) and significantly lower in the Ca^{2+}-free medium group (66.2 ± 13.95 cells) compared with the LIPUS-only group (180.53 ± 13.67 cells; Fig. 6c). Therefore, we concluded that the neurite outgrowth and proliferation of NSC-34 cells were affected by LIPUS-induced Ca^{2+} signaling.

3.4. Ca^{2+}-dependent transcription factors by LIPUS

Transcription factors, such as NFAT, NKκB, and CREB, are activated in Ca^{2+}-dependent responses. These transcription factors affect neural morphology by regulating dendritic growth and branching and spike formation [20,50,51]. In this study, the NSC-34 cells were treated with a Ca^{2+}-free medium, La^{3+}, and Gd^{3+} 30 min prior to LIPUS stimulation. No significant nuclear translocation of NFAT and NFκB occurred without LIPUS stimulation in the control group and the Ca^{2+}-free-medium-, La^{3+}-, and Gd^{3+}-treated groups. However, LIPUS stimulation induced the nuclear translocation of NFAT and NFκB in a manner similar to that observed among the positive control groups. Furthermore, the LIPUS-induced activation and nuclear translocation of NFAT and NFκB can be inhibited by Ca^{2+}-free media, La^{3+}, and Gd^{3+} (Figs. 7a, 8a). For NFAT, the nucleus-to-cytosol ratios (N/C ratios) of the positive control (ionomycin), control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups without LIPUS stimulation were 1.03 ± 0.04, 0.51 ± 0.02, 0.49 ± 0.02, 0.5 ± 0.09, and 0.47 ± 0.09, respectively, whereas the N/C ratios of the control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups with LIPUS stimulation were 1.15 ± 0.08, 0.64 ± 0.03, 0.62 ± 0.11, and 0.58 ± 0.10, respectively (Fig. 7b). For NFκB, the N/C ratios of the positive control (PMA), control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups without LIPUS stimulation were 0.86 ± 0.15, 0.35 ± 0.04, 0.41 ± 0.06, 0.37 ± 0.01, and 0.39 ± 0.09, respectively, whereas the N/C ratios of the control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups with LIPUS stimulation were 0.91 ± 0.13, 0.46 ± 0.06, 0.44 ± 0.07, and 0.46 ± 0.11, respectively (Fig. 8b). In contrast to the other transcription factors, almost all of the cells' CREB was localized in the nuclei. The LIPUS stimulation and pre-stimulation of the Ca^{2+}-free medium, La^{3+}, and Gd^{3+} affected the distribution of CREB (Fig. 9). Therefore, we used the

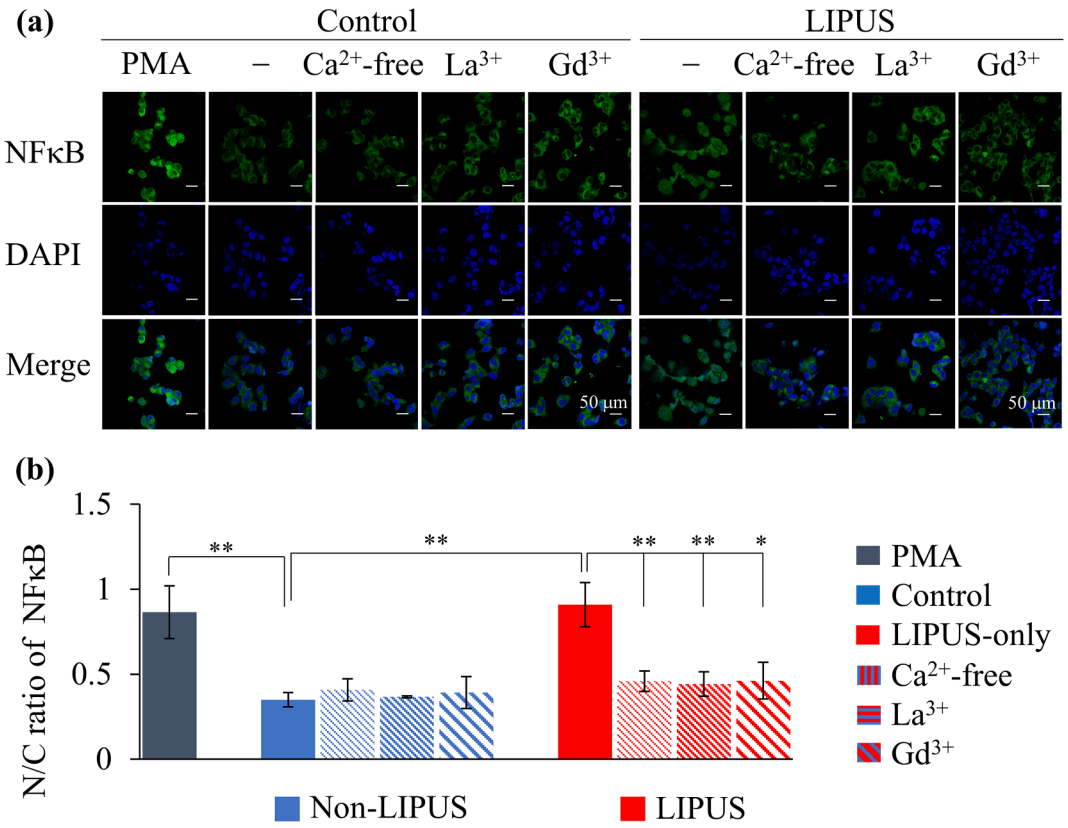


Fig. 8. LIPUS stimulation induces activation of transcription factor NFκB in NSC-34 cells. The 5 × 10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. Ca^{2+} channel blockers (200 μM La^{3+} and 50 μM Gd^{3+}) and a Ca^{2+}-free medium were administered for 30 min before LIPUS stimulation. Cells in the positive control groups were treated with 1 μM PMA for 30 min. (a) Immunofluorescence staining of NFκB (green) and nuclei (blue); images were taken using confocal microscopy. Scale bar: 50 μm. (b) Quantitative analysis of the nucleus-to-cytosol ratio for NFκB fluorescence intensity. Each bar represents the mean ± SEM from three independent experiments (5 pictures were taken for each condition in every experiment). *$p < 0.05$; **$p < 0.01$.

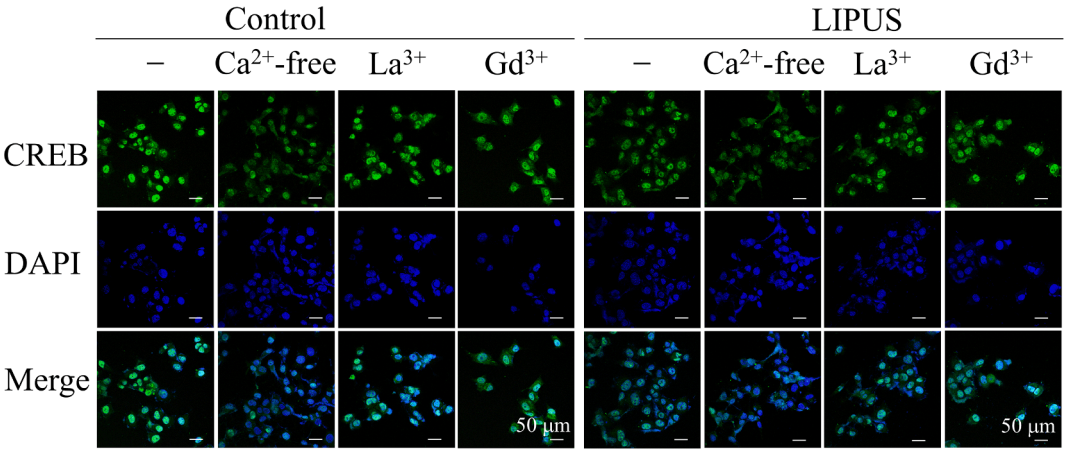


Fig. 9. Transcription factor CREB was not activated during LIPUS stimulation. The 5 × 10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. Ca^{2+} channel blockers (200 μM La^{3+} and 50 μM Gd^{3+}) and a Ca^{2+}-free medium were administered for 30 min before LIPUS stimulation. Immunofluorescence staining of CREB (green) and nuclei (blue); images were taken using confocal microscopy. Scale bar: 50 μm.

phosphorylated CREB (p-CREB) as a marker for activated CREB in the nuclei. No significant differences in p-CREB were identified among the groups (Fig. 10a). As indicated in Fig. 10b, the mean integrated densities of the control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups without LIPUS stimulation were 17.09 ± 1.54, 17.61 ± 0.68, 17.10 ± 1.43, and 17.69 ± 2.45, respectively, whereas the mean integrated densities of the control, Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups with LIPUS stimulation were 18.54 ± 0.97, 18.38 ± 0.46, 18.23 ± 0.85, and 17.46 ± 1.63, respectively. These results indicate that the NFAT and NFκB were activated by LIPUS stimulation, but CREB was not.

3.5. *LIPUS cytotoxicity on NSC-34 cells*

The cell-impermeable DNA dye Ethidium and permeable DNA dye

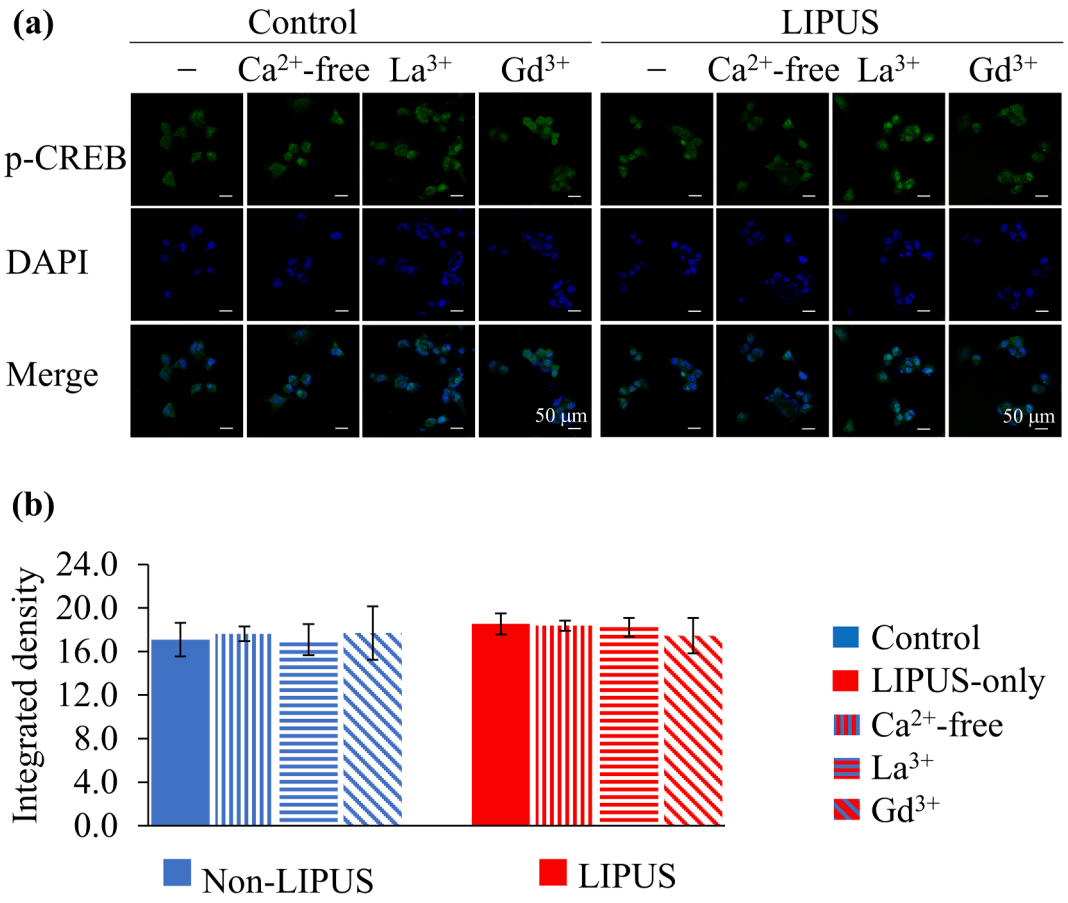


Fig. 10. Transcription factor phosphorylated CREB was not activated by LIPUS stimulation in NSC-34 cells. The 5×10^4 NSC-34 cells seeded in 3-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (0 mW/cm^2) LIPUS on days 1 and 3. Ca^{2+} channel blockers (200 μM La^{3+} and 50 μM Gd^{3+}) and a Ca^{2+}-free medium were administered for 30 min before LIPUS stimulation. (a) Immunofluorescence staining of phosphorylated CREB (p-CREB; green) and nuclei (blue); images were taken using confocal microscopy. Scale bar: 50 μm. (b) Quantitative analysis of mean integrated density value for p-CREB. Each bar represents the mean ± SEM three independent experiments (5 pictures were taken for each condition in every experiment).

Hoechst 33,342 were used to investigate whether LIPUS induces cytotoxicity in NSC-34 cells. The cells that did not undergo LIPUS treatment comprised the control group. On day 4, no notable difference in NSC-34 cell viability was identified between the control group (dead cells: 2.2% ± 0.3 %) and the LIPUS group after LIPUS stimulation (dead cells: 2.4% ± 0.3%; Fig. 11). Thus, our findings suggest that LIPUS simulation does not harm the viability of NSC-34 cells.

4. Discussion

Ultrasound has been used widely in clinical and pre-clinical diagnoses for various diseases [52–54]. Besides the diagnosis, therapeutic ultrasound has also played a very important role in clinical applications recently. Several studies have demonstrated that protecting cell survival is key to successful stimulation therapy for ALS patients [55–58]. In addition, the ability of LIPUS to stimulate cell growth and differentiation has been reported [30,59–65]. In this study, NSC-34 was selected as a hybrid cell line to investigate the effect of LIPUS. These cells exhibit the properties of a motor neuron when subjected to differentiation and maturation protocols. NSC-34 cells have high homogeneity in type, size, or response to stimuli, and can be cultured for a long time and continue to proliferate. In fact, NSC-34 cells comprise the most stable line of motor neurons and are widely used in models of MND pathology [12,55,56] and characterized by the expression of certain proteins specific to motor neurons [47,66,67]. Thus, the NSC-34 cell line is a valid model to investigate the increase in Ca^{2+} levels, which is key to signaling that induces transcription factor activation, supporting proliferation and altered neuronal morphology [51].

In this study, we demonstrated that LIPUS exerts a positive effect on growth, differentiation and Ca^{2+}-dependent transcription factor activation in NSC-34 cells. We determined that the optimal parameters for LIPUS stimulation of NSC-34 cells are the intensity of 357 mW/cm^2 for 8 min on days 1 and 3 (Fig. 4). A high proportion of the NSC-34 cells were incorporated with EdU to examine DNA replication during cell proliferation under the previous conditions (Fig. 4). Under these conditions, LIPUS significantly increased the level of morphological changes of mature NSC-34 cells, such as the length of neurite outgrowth, the percentage of neurons undergoing nerve processes, the extent of cell spreading, the number of attachment points, and the number of end points of branches (Fig. 5), as well as the viability of NSC-34 neurons (Fig. 11). This further indicates that LIPUS can protect and enhance the viability and development of NSC-34 motor neurons.

In addition, Ca^{2+} signaling regulates the cytoskeleton to control growth cone *morphology* and motility during axon guidance. Cell morphology mainly depends on the construction and dynamics of the cytoskeleton, especially actin filaments and microtubules. Ca^{2+} controls the formation and activity of the cytoskeleton, usually by modulating an associated motor protein or one that affects the structural organization of the polymer. Many cellular functions are related to the regulation of the cytoskeleton by Ca^{2+}, such as plasma membrane protrusion, actomyosin contraction, cell migration, chemotaxis, mitosis and cytokinesis [16–20]. The morphological changes of the NSC-34 cells were observed under LIPUS stimulation. A Ca^{2+}-free medium and two Ca^{2+} channel blockers, La^{3+} and Gd^{3+}, were used to examine the LIPUS-induced Ca^{2+}

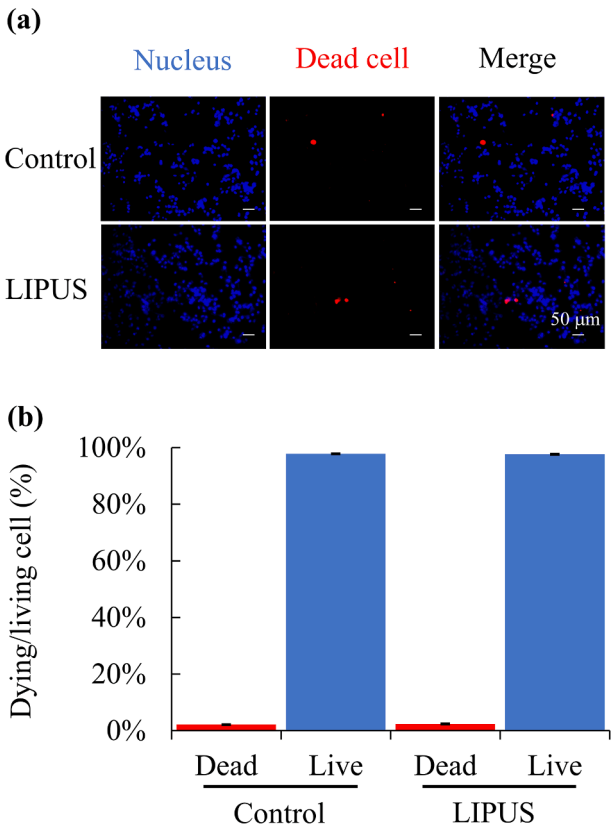


Fig. 11. LIPUS stimulation without cytotoxicity on NSC-34 cells. The 5×10^4 NSC-34 cells seeded in 3.5-cm plastic dishes for 24 h were stimulated for 8 min with (357 mW/cm^2) or without (control group: 0 mW/cm^2) LIPUS on days 1 and 3. On day 4, the NSC-34 cells were incubated for 30 min with Ethidium (red) for dead cells and Hoechst 33,342 (blue) for all nuclei. (a) Fluorescence images were taken under an inverted fluorescence microscope. Scale bar: 50 μm. (b) Quantitative analysis of the percentages of live and dead cells with or without LIPUS stimulation. Each bar represents the mean ± SEM from three independent experiments (5 pictures were taken for each condition in every experiment).

signaling on cell morphology. As shown in Fig. 6, neurite outgrowth was observed among fewer of the NSC-34 cells in the Ca^{2+}-free medium, La^{3+}, and Gd^{3+} groups than of those in the LIPUS-only group. The mean neurite length of the control LIPUS group was significantly longer than those of the groups with inhibited Ca^{2+} signaling. Therefore, Ca^{2+} signaling is involved in LIPUS-induced morphological changes of NSC-34 cells.

Moreover, Ca^{2+} signaling with different spatial–temporal LIPUS characteristics, including amplitude, frequency, duty cycle, and duration, are important for determining cell fate [68–70]. Different types of cells and molecules respond to different frequency and amplitude ranges of Ca^{2+} oscillation waves in vitro and *in vivo* [71]. Ca^{2+} signaling is most commonly used to regulate almost 60–70% of genes in the human genome through three major Ca^{2+}-dependent transcription factors including NFAT, CREB, and NFκB [72]. The regulation of dendritic growth with Ca^{2+} influx via VSCC is directly related to Ca^{2+}-dependent transcription. The most characteristic target is the transcription factor CREB, but the activation of CREB is insufficient to induce dendritic growth [16]. Therefore, activation of other Ca^{2+}-dependent transcription factors that contribute to the proliferation, morphological changes and activation of nerve cells [17]. Extensive research has verified the beneficial role of calcineurin/NFAT signaling in the survival, proliferation, and differentiation of nerve precursor cells through Ca^{2+} influx through L-type channels and NMDA receptor–dependent Ca^{2+} influx [23,73]. Furthermore, morphological development is determined by active TrkB-Akt and Ca^{2+}-CaMK signaling pathways [21,37,74] and by the GSK-3β/β-catenin/cyclin D1 signaling pathway [32], as demonstrated in some LIPUS studies.

It has been reported that p-CREB could be activated depending on the opening of Piezo1 in HEK293T cells by ultrasound stimulation [75]. In our study, CREB or p-CREB was already present in the nucleus and was phosphorylated in an activated state before LIPUS stimulation (Figs. 9 and 10). LIPUS stimulation did not affect the distribution of CREB in the nucleus, indicating that CREB continued to be active. However, LIPUS stimulation could significantly promote the nuclear entry of NFAT (Fig. 7) and NFκB (Fig. 8) to enhance the transcriptional activity. This may explain the finding in this study that LIPUS stimulation can promote the proliferation (Fig. 4) and morphological differentiation (Fig. 5) of NSC-34 cells. The study by Lai et al. using optogenetics to regulate different Ca^{2+} signaling patterns also confirmed that Ca^{2+} oscillations at different ultrasound frequencies and duty cycles have different effects on the activation of Ca^{2+}-dependent transcriptional factors such as NFAT, NFκB, and CREB [18]. Therefore, Ca^{2+} signaling affects neuronal morphology by regulating dendritic growth and branching and spike formation, and the regulation of dendritic growth by Ca^{2+} flux is directly related to Ca^{2+}-dependent transcription. Different frequencies, intensities, or durations of ultrasound stimuli may cause different Ca^{2+} signals and effects in different cells.

5. Conclusion

In summary, this study aimed to investigate the effect of LIPUS on the development of the NSC-34 motor neurons and its potential underlying mechanisms. Our results indicate that LIPUS therapy (with a frequency of 1.15 MHz, I_{SPTA} of 357 mW/cm^2) can significantly increase the number of cells, DNA replication, mean length of neurite outgrowth, and level of several factors related to the morphological changes in NSC-34 cells. These effects of LIPUS on neurodevelopment are attributed to the activation of transcription factors NFAT and NFκB. Furthermore, the blockage of Ca^{2+} channels can suppress the stimulating effect of LIPUS on neural morphological development. Therefore, further studies are required to investigate the effects of LIPUS on nerve regeneration in animal models and the molecular mechanisms. These findings jointly provide key insights into how LIPUS can promote the development of neural changes through Ca^{2+} signaling to activate Ca^{2+}-dependent transcription factors. Therefore, LIPUS is potentially useful in noninvasive stimulation therapy for MNDs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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